Proposal for Riget Zoo Adventures (RZA)

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Business context

Our client, Riget Zoo Adventures (RZA), is a well-established business that is interested in digitalisation in order to keep existing customers satisfied as well as engaging new clients, Currently RZA offers a safari-style wildlife zoo, an on-site hotel and educational visits. A client is looking for a digital solution that would information about attractions and facilities, materials to support educational visits and allow users to book tickets and stays in the hotel.

I am proposing to develop an interactive website that would include all the features that the client requires and some more. In the following passages I will be explaining and justifying my proposal.

Empathy map

Says	**Thinks**
- I need more information before visiting this zoo. - Could I buy a book that has all the animals description, so I get to learn while I'm on my educational trip as well? - Could I book my tickets and book a hotel in advance via website? - Could I create my account so I could manage my bookings? I need to cancel my booking for this Saturday because my child is not feeling well.	- I wish there was a website where I could see all the animals, it would be easy to decide if I want to go this zoo - It would be so much easier if they just had a booking system and I could book a ticket in advance instead of standing in the line for hours. - I would rather book a hotel in advance, what if there are no rooms left when we visit. - Oh, I wish I had more personalised experience, I wish I had my own account. I don't want to lose my money just because I couldn't cancel my visit.
Does	**Feels**
- Comes to the zoo with very little awareness - Comes to visit, but the zoo is too full today - Comes to visit and finds out that the hotel is overbooked	- Overwhelmed - Confused - Stressed and annoyed because of the que to by ticket - Willing to be reassured that the hotel is booked for the time they come

User stories

As a [user]…	I want …	So that …	Acceptance Criteria
First time customer	To visit a website and find some information on what the zoo is offering.	I can compare it with other zoos and see why I should choose this one, as well as that I would be able to see if	The system should allow user to see the clear navigation bar at the top, in the main section there should be

	Book my tickets quickly online.	there are animals that I am particularly interested in. I will not have to wait in the que for tickets, when I visit.	a carousel in the main section that will prompt users to all the other sections. When user visits the about page, all of the necessary info will be displayed in there such as how to get here, opening times, animals, facilities, hotel info. On the top right corner there will be a “Book” button that will prompt user to the booking page, it could be a ticket booking or a hotel booking.
Student	To visit website that is easy to follow and does not have unnecessary information, I also want to find educational trip section on there where I can browse and get all necessary materials for my trip, for ex book that describes all the animals that I’m going to see in the zoo.	When I am on my trip I was fully aware of what is going on and had enhanced experience, I also want to learn more when I’m on the trip.	The system should provide the educational visits page for students. On this page there should be a catalogue of things that they might need while on the trip.
Parent	To book a hotel and tickets in advance	I’m sure that when I come with my children we will have where to stay, and I will not need to drive all the way back home the same day.	In the booking area there is an option to book a hotel.
Loyal customer	To have a discount when I visit for the second time. To have my account where I would see how when I have visited. And to be able to cancel my future bookings if I need to do so.	If I visit multiple times, I can save some money. I also want to feel a more personalised experience.	The website should contain an option to create an account and login. When user is on any of the pages, he/she should be able to easily navigate to the Account section. When on the Account page, user should be able to Manage bookings, have reward for each visit. See his/her previous visits.

Functional Requirements

Homepage

The system should display homepage that consists of Navigation bar, main section and footer. In the main section website provides A carousel of pictures with links that prompt user to other areas of the website, ex about page.

No data inputs required for this page.

When the user clicks on the links to different pages it should go to the required page. Ex. User clicks on the Book button, it should quickly navigate the user to the booking page.

Navigation Bar consists of:

- Home
- About
- Educational visit
- Book
- Account

About Page

The system displays the about page which is similar to the homepage and consists of the navigation bar, main section: Opening times, Our animals, Hotel info, How to get here; and a footer.

No data inputs required for this page.

Educational visits page

The system displays the educational visit page which consists of the navigation bar, main section where the user can browse the catalogue of useful materials for the educational trip and the pricing of it, as well as that user should be able to download a pdf guide with all the animals that are in the zoo.

No data inputs required for this page.

Booking system

The system should have two services options to book: book a room or book a ticket. Then user the correct form should be displayed, first user is prompted to select a day and timeslot, then fill in personal details. After user fills in the form, order should be placed and sent to the database, user should see the message of the successful placement or in case of any errors a message that something went wrong during ordering.

User input requirements:

Booking a room	Booking a ticket
•Stay from [], to [] •Room type (Single room, Double Room, Twin Room, Double Room - Disability Access) •Number of visitors •Full name •Email	•Date and time •Type of ticket (Adult,Child, Student, Educational trip) •Full name •Email

Personal Account

The system should allow user to create an account, or sign in, if the account already exists. Once signed in to their account user should be able to manage their bookings ex. cancel, user should also see previous stays and rewards. The system should allow user to log out once needed.

User input requirements to create an account:

User input requirements to login into the account:

Non-functional Requirements

Security

The system should be secure and robust:

- System must validate all data inputs.
- System must use secure cookies.
- System must use SHA256 Encryption for the passwords.

Maintainability

It is important that after deployment system can be maintained cost-effectively.

To ensure good maintainability, I will only use good coding practises throughout the code, I will ensure that naming conventions are relevant and easy to follow, I will only use standard API formats.

Performance

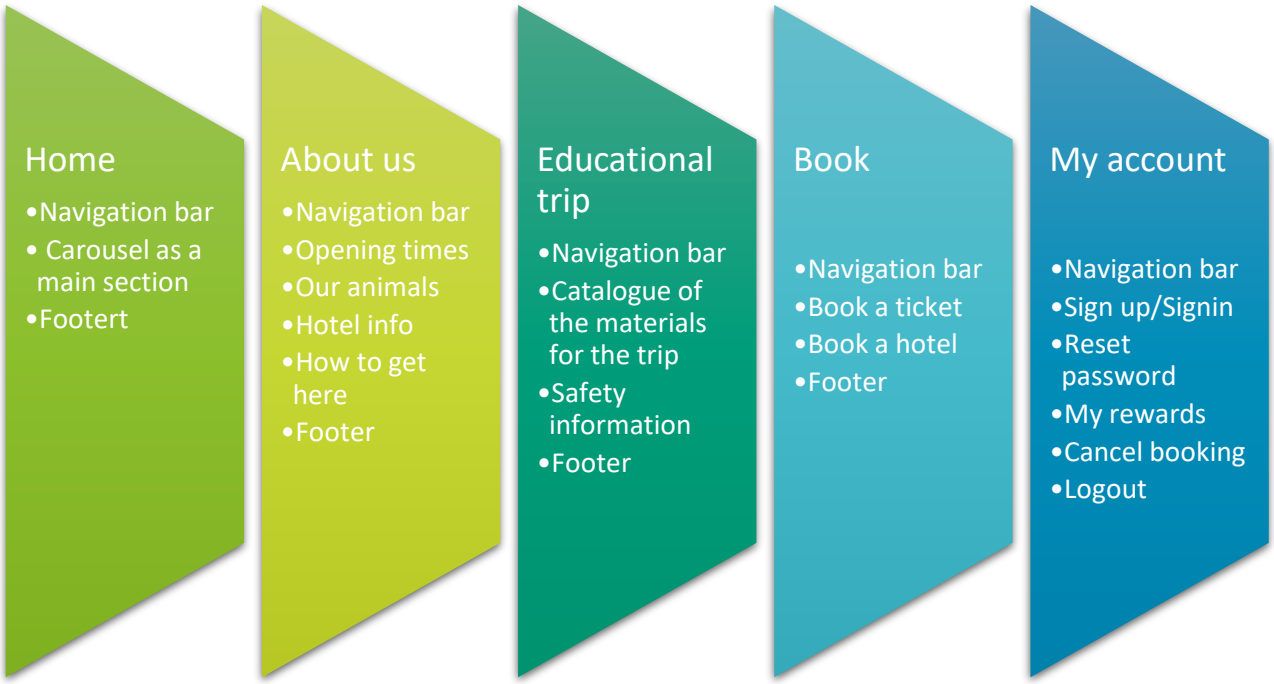
- Page speed/load: webpage should be loaded within 2-6 seconds.
- Latency: it should take not more than 100 milliseconds for system to respond to the user.
- The system should not be overloaded with API calls, as they might slow down the performance.

Usability -> Accessibility

The system should be straightforward to use, both for the end-users and the content editors that oversee backend system.

System should be adapted to different user needs: the system should be following Web Content Accessibility (WCAG) guidelines.

Decomposition



KPI's

Business KPI's

Trustpilot reviews

the higher the ranking on the third party's ranking websites such as Trustpilot the better it is for the business, the reason being is that usually people look for the best option and it is crucial for them to be able to see what other people that have already visited place think, it is due to complex human psychology. Recent example: to conduct research and make a valid proposal I first had a look on the Trustpilot to see what competitors offer.

Visitor satisfaction rate

it can help identifying areas of improvement and therefore boost customer's experience. One way to track this KPI is to ask customers to fill in short surveys. Industry standard is to keep this rate 80% and higher, however the higher rate, better it is for the business, meaning that everything is managed to the highest standard. One drawback is that sometimes customers are too busy to do it and some of the are not bothered to read the question properly. To mitigate that, we need to make sure that the questionnaire is short and has and engaging element to it.

The number of educational programs offered

this can allow the business to make strong connections with public and have even more customers. It can be calculated by adding all of the programs offered per year. If the number growth over years that would mean that business has an adequate scaling up.

Percentage of repeat visitors

it shows us the business' ability to maintain customers. Therefore, if there's a drop in this percentage business can easily spot that there is a problem with customer satisfaction and thus perhaps, they should consider their customer service more. This KPI will allow business to identify areas for improvement and faster long-term loyalty. In order to calculate this KPI business needs to have the number of repeat visitors and total amount of visitors. Then divide repeat visitor by total visitors and multiply by 100.

Revenue growth rate

asses RZA's financial performance, which can help to perform better informed decisions. The higher the business revenue growth rate the better.

Website KPI's

Page Load time

It is essentially how long it takes for page to load and appear on the user's screen. It is crucial that this time is as low as possible. Even few seconds can result into the loss of the protentional client. With the development of the digital sector, people tend to wait less and less if the there are any troubles accessing website as they are well aware of the vast amounts of options they've got.

This KPI can be seen using reporting tool, for example Google Analytics.

Bounce Rate

This KPI shows the percentage of users that visit the website and then leave immediately without taking any actions. It is logical that if this rate is high the website is either performing poorly or it is simply not appealing well to the user, one other reason might be hidden in the advertisement, if the website is offered to the wrong audience, it might be that they click without any interest in the sector at the first place. A good website though will keep the visitors to interact at least with few pages before leaving. Therefore, it is a crucial tool that will allow to see if the website needs any changes or not.

Traffic by source

It allows to see different sources of traffic. For example, what devices are most used to visit the website, and therefore RZA could investigate farther development of that device. If let's say most traffic comes from smartphones it is sensible to develop a mobile app.

Conversion rate

It indicates how well the website prompts customers to complete the action desired by the client. For example, RZA wants to have n number of bookings and y amount of account registrations per day then we would need to take those two goals per specific amount of time and divide it by total number of users and multiply it by 100%.
It is important that this rate is reasonably high, if the rate is low that would mean that the website is doing its function very poorly.

Average time on page

It shows us how much time on average user spends on the website. Depending on the page this time can indicate different outcomes. For example, if the average time on the homepage or about page is high it's a good indicator as it shows that the content is engaging and relevant for the user however if KPI is high on the booking page then that might show that there are some errors or it's to hard for the user to understand. And thus, some changes should be made.

A description of proposed solution

In the above paragraphs I was decomposing, analysing, and setting the requirements for the system. In this section I will be summing up the proposal and its main benefits to the client – RZA.

The final proposal is an interactive website that will include following

Key features and benefits:

- Inclusive solution

  The website is going to be an inclusive platform that will allow all users navigate through and use its functionality. In this way RZA will ensure that all customers fill included and therefore satisfied with their services.

- Quick and easy booking

  One of the most important features is booking a ticket or a room in advance, which will help reduce ques for the tickets kiosks in the Zoo, as well as that users will be resting assured that they will have the place to stay overnight, not worrying that the hotel is overbooked.

- Materials for educational visits

  The website will allow students and their parents to see all the necessary information and materials for educational visits. As well as that they would be able to download a pdf with all the animals in the Zoo.

- Info Blocks

  Will provide customers with information about the zoo and facilities, opening times, hotel, and animals

- Personalised experience

  For loyal customers website offers to create an account and have some awards for each visit, for example a discount coupon. As well as that, customers will be able to see their previous visits and manage their current booking. Which will ensure that customers receive personalised experience and come back to the RZA.

Justification

In the above paragraphs I have proposed a digital solution to the client. Which is an interactive website that meets different users needs, I have also stated the requirements: functional and non-functional and provided decomposition. I this section I will be justifying my proposal.

How the recommended solution meets the needs of the client.

Inclusion-driven solution

One of the client's needs is to support a wide range of customers, therefore a lot of emphasis will go into making sure that the website is built with all the customers in mind. The frontend will be developed based on the reliable Bootstrap templates, that follow WSAG. All the colours and contrasts will be checked before implementing the design into the protype.

The reason being we want the website to be accessible for everyone, ensuring that all the customers are having a good experience. As well as satisfying existing customers, website will be a powerhouse for engaging new customers.

Efficient way to buy and book

Another client requirement is to allow users to book tickets and hotel rooms online. The website will contain an easy to navigate booking system with a robust backend and database. This will help our client to keep track of all the tickets sold and rooms booked. Client will be able to analyse this data and make a better informed strategical and technical decisions. On top of that customers satisfaction rate will grow as they don't have to waste time in the ques and worry about availability of rooms in the hotel.

All about educational visits section

Another client need was to support educational visits. I'm proposing to make a separate page dedicated to the educational trips where all the necessary information will be displayed. In this way it will be easy to spot on the navigation bar. And this will help to enhance students' experience.

Informational blocks

Another feature that client required is to do with information about attractions and facilities. What I have proposed is to make an about page where end-user could find all the necessary information. After research into industry, I have found that most of the zoo websites have the animals' sections, where users could see all the current animals in the zoo.

I think it is a reasonable feature to add as it will help customers to make more informed decision. Perhaps, RGA has exactly the animal that one of the protentional clients wants to see, this would bring a new customer. However, if the opposite happens and the customer is interested in the particular animal, but that animal is not in the Zoo, then it would be sensible to the user not to come, which is also a benefit for RZA, as this customer will have a very high chance of coming out not satisfied, which could lower RZA ranking on the third-party rankings' website.

As well as animals' section, I have proposed to add information about the hotel, opening times and attractions. Which will keep protentional customers informed.

Personalised user experience

As an additional feature client has mentioned that users could have their own accounts and perhaps a loyalty system. What I have proposed is a robust registration / login system. Users will

be able to create an account and have some awards for each visit, for example a discount coupon. As well as that, end-users will be able to see their previous visits and manage their current bookings. This will improve their experience with RZA and therefore promote brand's reputation. In the long-term allow RZA for scaling up.

How protentional risks will be mitigated.

In the proposal above I have mentioned that the website will contain following features: Info page (About Page), Educational visits, Booking system and personalised account system with loyalty program. In this passage I will walk through the potential risks and their mitigations.

Informational inaccuracy (Poor data quality)

This could lead to lost revenue, poor customer experience and in long term even to the reputational damage.

The way to mitigate it is to hire staff that will oversee editing content and making sure it is up to date. Although for the scope of this project I will not be implementing CMS (Content Management System), it might be a sensible solution for farther development, as it will reduce the complexity of maintaining the website, therefore increase efficiency. The reason being is that it allows for content change in a nice user-friendly interface, which essentially does not require programming knowledge, therefore could be done by content creators.

Miscommunications with schools and students

when it comes to educational visits can be an issue, when not handled gracefully. One example could be arranging a visit for the wrong time slot. Another could be misleading information about materials required on the web page. This frankly leads us back to the information inaccuracy issue and the way to mitigate those would be quite similar, however I would highly suggest that there are delegated to manage educational programs. In case of, something going wrong, they would be able to quickly react to the circumstances and come up with a solution.

Booking system – data loss

There are a handful of risks when it comes to the booking system and its implementation like overcomplicating the booking process, not having enough customer support, failing to customize accordingly to business needs. However, what I want to concentrate is system glitches and data loss which could be dreadful to the business reputation. For example, if the data loss happens and the customers that booked a hotel to stay the night, it would be extremely unpleasant experience to them having to change their place and looking for different place on the day of a visit.

To mitigate this, the system must be thoroughly tested. However, for data loss it's not always sufficient, therefore data backups will be a reasonable solution to this potential risk.

Personalised account – data breaches

One of the risks with the most consequences is a data breach. As for the login system, we will gather some customer information, there is always a risk of this data to be stollen. To mitigate this, company should have a robust security plan, that comprise of multiple prevention technics. For this scope I will be implementing password encryptions to enhance the security.

Regulatory guidelines and legal requirements will be addressed.

To make sure the solution that I have proposed above meet the low, I will ensure that the website is compliant with such legislations as GDPR (General Data Protection Policy), COPPA(Children's Online Privacy Protection Act), EU Cookie Law.

technology has evolved beyond mere directional guidance, offering companies and users a comprehensive and dynamic experience. These interactive mapping software solutions incorporate real-time data, satellite imagery, and machine learning algorithms to provide accurate and up-to-date maps. Businesses can use these interactive tools to customize their maps to meet the needs of their audiences.

Whether used for urban planning, logistics optimization, or travel navigation, mapping software has transcended traditional boundaries. It not only enables users to find the quickest route from point A to point B but also facilitates a deeper understanding of geographic landscapes. With features like augmented reality overlays and interactive layers, mapping tools continue to redefine our spatial awareness, making them an essential companion in our daily lives.

Augmented Reality (AR) for Enhanced Experiences

For travel and tourism companies aiming to remain competitive, the integration of augmented reality into customer experiences is essential. The AR market is projected to experience a remarkable CAGR of 38% by the year 2030. This growth forecast underscores the increasing importance of AR in the industry, highlighting its pivotal role not only in enhancing customer engagement but also in securing a competitive edge for businesses within the travel and tourism sector.

In 2024, we can expect AR to become an integral part of travel apps, providing users with real-time information and enhancing the experience of their surroundings. Imagine walking through historical landmarks and using AR to witness ancient civilizations coming to life or getting instant translations of street signs. AR will bridge the gap between the physical and digital worlds, offering tourists a more immersive and interactive journey.

Blockchain for Secure and Transparent Transactions

Blockchain technology is gaining traction in the tourism industry, offering secure and transparent transactions. By 2024, blockchain will likely become widely adopted for booking accommodation, flights, and travel services. Its decentralized nature ensures transactions remain tamper-proof while simultaneously lowering fraud risk.

Smart contracts will automate processes, ensuring seamless and secure interactions between travellers and service providers, ultimately fostering trust in the industry.

Artificial Intelligence (AI) for Personalized Travel Experiences

Artificial intelligence is becoming increasingly sophisticated in tailoring travel experiences to individual preferences. In 2024, AI-powered travel assistants will be the norm, providing personalized recommendations for accommodations, activities, and dining based on travellers' preferences and past behaviour.

Machine learning algorithms will continuously refine suggestions, ensuring that each trip becomes a unique and tailored adventure. AI will also play a crucial role in customer service, with chatbots providing instant assistance and support throughout the travel journey.